Domain Adaptive Remote Sensing Scene Recognition via Semantic Relationship Knowledge Transfer

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***Abstract*— Scene recognition has attracted rising attentions of many researchers in the remote sensing fields, owing to the rapidly advancing of remote sensing devices in recent years. However, images obtained from various sensors dominate diverse sensor-specific characteristics, which will dramatically weaken the model transferability trained on a source data domain to a different target domain on account of the domain shift issues. To mitigate the domain discrepancy, most existing methods attend to align the cross-domain distributions. While the valuable knowledge of semantic relationships between different scenes is generally overlooked, and the underlying correlation across scenes cannot be fully discovered. For the sake of tackling this challenge, we propose an adaptive remote sensing scene recognition network, which can successfully transfer both the discriminative knowledge and cross-scene relationship from source to target. Specifically, in this article, we acquire sensor-invariant representations in an adversarial manner and realize fine-grained conditional distribution alignment contrastively. In such a way, the tremendous domain gap can be mitigated to a large extent, and the discriminative and well-matched representations will be derived favorably. In addition, we explicitly construct classwise relationship distributions belonging to two domains, respectively, and minimize their divergence to conduct semantic relationship knowledge transfer (SRKT), for the purpose of sufficiently unearthing the intrinsic semantic relative structures that can prompt generality of the model in the target domain. Finally, we conduct multiple experiments on representative multidomain remote sensing benchmarks, and the extensive experimental results demonstrate the superiority of our proposed approach.**

***Index Terms*— Domain shift, remote sensing, scene recognition, semantic relationship knowledge transfer (SRKT).**

I. Introduction

CURRENTLY, remote sensing scene recognition, which aims to classify the category of remote sensing images, has been widely utilized in numerous application areas, such as vegetation management [1], urban planning [2], and environmental monitoring [3]. Generally speaking, remote sensing images can be roughly divided into two categories according to the difference of sensors, including satellite images and aerial images [4], [5]. Benefiting from the rapid development of deep learning in recent decades, remote sensing scene recognition gains continuously excellent achievements in various scenarios [6]. Admittedly, the impressive performance of supervised deep learning usually relies on extensive annotated samples [7], [8]. However, labeling massive data are labor-expensive and time-consuming. In addition, data collected from diverse domains also bring domain shift/gap issues inevitably [9], [10], [11], [12]. Therefore, how to make full use of the existing labeled dataset to gain a well-trained model for generalizing to completely new and different data is an attractive subject for human beings. To tackle the above challenge, unsupervised domain adaptation (UDA) is a popular technique, which can effectively narrow the cross-domain discrepancy and transfer valuable knowledge across source and target. Also, many pioneers have made fruitful attempts in the past decades [13], [14], [15], [16], [17], [18], [19].

In this article, we focus on the challenging cross-domain remote sensing scene recognition tasks. Actually, remote sensing data obtained by various remote sensors dominate diverse sensor-specific characteristics, which will dramatically weaken the model transferability trained on a source data domain to a different target domain on account of the domain shift issues [20], [21]. Obviously, the resolution divergence and imaging condition difference caught by distinct remote sensors will produce inevitable heterogeneities in cross-domain remote sensing images, which is the main obstacle we should overcome for the cross-domain remote sensing recognition tasks [22], [23].

Fortunately, several works carry out meaningful explorations and achieve promising results in remote sensing scene recognition [24], [25], [26], [27], [28], [29]. To name a few, Lin et al. [24] propose a dual-adversarial learning method (DuEDL) that conducts two adversarial learning phases to transfer discriminative knowledge from source to target. Lin et al. [25] propose a multilabel aerial image classification model based on domain adaptation (DA-MAIC) that leverages the graph convolutional network (GCN) to predict multiple image labels and the domain classifier to extract domain-invariant representations for mitigating the domain shift. To exploit the classification information in the target

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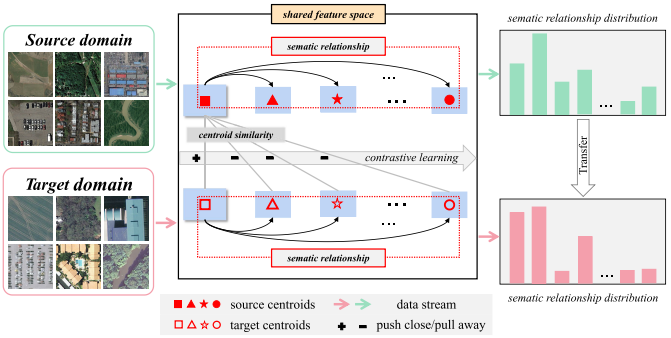


Fig. 1. Motivation of our proposed SRTK method, illustrating feature conditional distribution alignment based on contrastive manner and SRKT (similarity between any two classes).

domain, Zheng et al. [26] propose a domain adaptation via a task-specific classifier (DATSNET) framework for high spatial resolution (HSR) image scene classification. For cross-domain remote sensing image semantic segmentation, Li et al. [30] present a novel objective function with multiple weakly-supervised constraints that can effectively reduce the disadvantageous influence of data shift.

Though impressive, these methods pay almost all attention to conducting feature distribution alignment or utilizing adversarial operation for extracting domain-invariant features for the sake of improving the transferability of model [31], [32], [33]. However, they ignore the crucial information on semantic relationships among different classes, resulting in the underdevelopment of intrinsic connection among different scenes, which is precious guidance for training models when targets are unlabeled. Note that Li et al. [34] consider the semantic relations in remote sensing scene categories. They propose to exploit the representation learning of remote sensing knowledge graph (RSKG) to improve the semantic representation ability of scene categories. By pursuing cross-modal matching between visual features and semantic representations via their proposed deep alignment network (DAN), they address zero-shot and generalized zero-shot remote sensing image scene classification. But they need domain experts to participate in constructing the RSKG. For better illustrating, as shown in Fig. 1, we select six common scenes from popularly used remote sensing images datasets Aerial Image Dataset (AID) [35] and UC Merced Land Use Dataset (UCM) [36], including: "farmland," "forest," "parking," "industrial," "residential," and "river." In "river" scene images, the forest content usually appears beside the river. So when seeing a "river," it is natural to associate it with associated scenes such as "river" and "forest," and rarely think about scenes such as "industrial" and "residential." In other words, the river is close to the "forest" and apart from the "residential" in a shared feature space. When this semantic relevance structure in the source domain is distilled to the target domain, we can improve the discriminability of data in the target domain. Inspired by this intuition, we model the semantic relationship distributions in two domains, respectively, for semantic correlation transferring and propose a simple and effective method to address the challenging domain adaptive remote sensing scene recognition problem.

In this article, we propose the semantic relationship knowledge transfer (SRKT) framework for remote sensing cross-scene recognition. Specifically, our proposed method consists of three main components: acquiring sensor-invariant representations in an adversarial manner, realizing fine-grained conditional distribution alignment contrastively, and SRKT. First, we make the target domain generator cooperate with the classifiers to yield domain-invariant features to eliminate domain shift. To be specific, we design two separate generators for the source and target domains, respectively, along with two shared but distinct classifiers for scene recognition in both domains. We employ the outputs of two classifiers to calculate the classifiers' prediction discrepancy and then maximize it. On the contrary, the features generated by the target generator will minimize the diversity. Through this adversarial strategy, the target generator produces features owning domain-invariant characteristics gradually. Next, the centroids of different classes constitute the classwise positive and negative semantic pairs. We exploit the contrastive operation to push the cross-domain centroids belonging to the same class closed and pull the centroids in different classes far away, achieving fine-grain alignment on the feature conditional distributions across domains, which allows the network to produce more discriminative and well-matched representations.

Furthermore, we independently model the semantic relationship distributions of two domains by calculating the similarity among classes and explore the effectiveness of SRKT from the source to the target domain. To mine the hidden connections among different scenes, we explicitly construct the semantic relationship between scenes in respective domains, generating two similarity distributions for each class. Finally, we minimize the discrepancy between distributions to transfer the semantic correlation knowledge effectively. Here, considering the target domain data has no labels, we assign the highly confident pseudo labels generated by the classifier to target domain data as labels. Contributions of this article can be summarized as follows.

1) A novel end-to-end trainable SRKT framework is proposed for domain adaptive remote sensing scene recognition, which yields domain-invariant features in an adversarial manner and more discriminative representations via contrastive alignment. Our proposed method could extract well-matched discriminative representations.
2) The SRKT focuses on transferring semantic relationship knowledge between the source and target domains to excavate inherent correlation among scenes, which can effectively promote the transferability of the model in the target domain. Compared with other related methods, SRKT can provide extra but crucial guidance for training models when the target domain is unlabeled.
3) For the sake of validating our proposed approach, we conduct comprehensive experiments on representative remote sensing datasets. The quantitative and qualitative experimental results demonstrate the superiority of our proposed method to the previous state-of-the-art methods.

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The rest of this article is composed as follows. The related works will be described in Section II. The details of the proposed SRKT framework are presented in Section III. Section IV illustrates the cross-sensor remote sensing datasets, experimental settings, experimental results, and related analysis. Section V concludes the whole article eventually.

II. Related Works

Since our method focuses on cross-domain scene recognition, we first investigate the popular UDA methods, and then describe the most relevant remote sensing scene recognition works based on UDA in current years.

A. Unsupervised Domain Adaptation

Feature distribution alignment usually is considered in UDA which draws the different feature distributions belonging to different domains close, to make data features of the source domain and target domain alike [9], [37], [38], [39], [40], [41], [42], [43], [44]. Wen et al. [45] propose a novel method for learning domain-invariant local feature patterns and jointly aligning holistic and local feature statistics. Through utilizing multirepresentation alignment, a multirepresentation adaptation network (MRAN) [46] is provided for cross-domain image classification task. Lee et al. [47] offer a sliced Wasserstein discrepancy (SWD) that can capture the diversity between the outputs of task-specific classifiers. Li et al. [48] present a simultaneous semantic alignment network (SSAN) to align the centroids for each category across domains and exploit correlations among categories simultaneously. To avoid the issues of sampling variability, class imbalance, and data-privacy concerns, Tanwisuth et al. [49] suggest a memory and computation-efficient probabilistic framework that extracts category prototypes and aligns the target domain features with them.

There also exist some UDA methods concentrating on adversarial strategy to produce domain-invariant features across source and target domains [15], [50], [51], [52], [53], [54], [55]. Ganin and Lempitsky [14] improve domain adaptation by making it hard for the domain discriminator to decide where the data belong and propose the gradient reversal layer (RevGrad), which is widely used now. Cao et al. [56] present a partial adversarial domain adaptation (PADA), which down-weighs the data of outlier source classes for training both source classifier and domain adversary and matches the feature distributions. Saito et al. [57] propose a novel adversarial strategy that maximizes the discrepancy between two classifiers outputs to detect target samples that are far from the support of the source, and learns a feature generator for generating target features near the support to minimize the discrepancy. Depending on adversarial dropout, Lee et al. [58] present a drop to adapt (DTA) to learn strongly discriminative features by enforcing the cluster assumption. Dai et al. [59] propose two transfer learning frameworks based on the multi-source domain adaptation methodology for sentiment analysis by combining the source hypotheses to derive a good target hypothesis. Shi et al. [60] propose a UDA framework based on progressive transfer for synthetic aperture radar (SAR) ship detection by transferring knowledge from the optical domain to the SAR domain.

Despite these UDA methods achieving excellent results in some scenarios, they are not specifically designed for remote sensing scene recognition. Because of remote sensing data obtained by various remote sensors dominate diverse sensor-specific characteristics, which conducts sharpen heterogeneity across domains. This dramatically weakens the effect on remote sensing cross-scene recognition. Therefore, to handle these tasks, the UDA method needs to be specifically developed considering the heterogeneity of the cross-domain scenes.

B. Domain Adaptive Remote Sensing Scene Recognition

Considering the remarkable achievement of UDA, lots of domain adaptive remote sensing scene recognition works have been proposed in recent years [1], [24], [26], [28], [61], [62], [63], [64], [65]. Lin et al. [24] put forward a DuEDL network that not only applies task-specific classifiers but also introduces domain-specific tasks for different domains during the adaptation process. Zhu et al. [1] come up with an attention-based multiscale residual adaptation network (AMRAN), in which both the marginal and conditional distributions are taken into consideration for more comprehensive alignment. Besides, the attention mechanism and the multiscale strategy are used to extract more robust features and complete information, respectively. The DATSNET framework is introduced [26] for HSR image scene classification, which conducts feature distribution alignment via task-specific classifiers and integrates it with minimaxing of the classifiers' discrepancy.

Though these methods partly overcome domain shift issues in remote sensing images and achieve excellent remote sensing cross-scene recognition performance, they overlook the potential semantic relationships among different scenes which are precious assistance for training models in the target domain. Motivated by this, we allow the target generator and the two communal classifiers to accomplish an adversarial procedure, yielding domain-invariant features to diminish domain shift. Further, we resort to the contrastive process to push the samples residing with the same class clustered and pull the samples in different classes far away, leading the network to produce more discriminative representations. More importantly, we model the semantic correlation distributions between different scenes of two domains and minimize their divergence to investigate the intrinsic semantic relationship structures helpful for the generality of the model in the target domain.

III. Method

Different from previous domain adaptive remote sensing recognition methods, SRKT emphasizes discovering intrinsic semantic correlation in the target domain for facilitating distinguished representation learning. We first aim to derive domain-invariant as well as discriminative representations by adversarial pattern and contrastive learning. Then the semantic prototypes generate two semantic relationship distributions for each class in the source and target domains, respectively. We reduce the discrepancy between them to explicitly transfer

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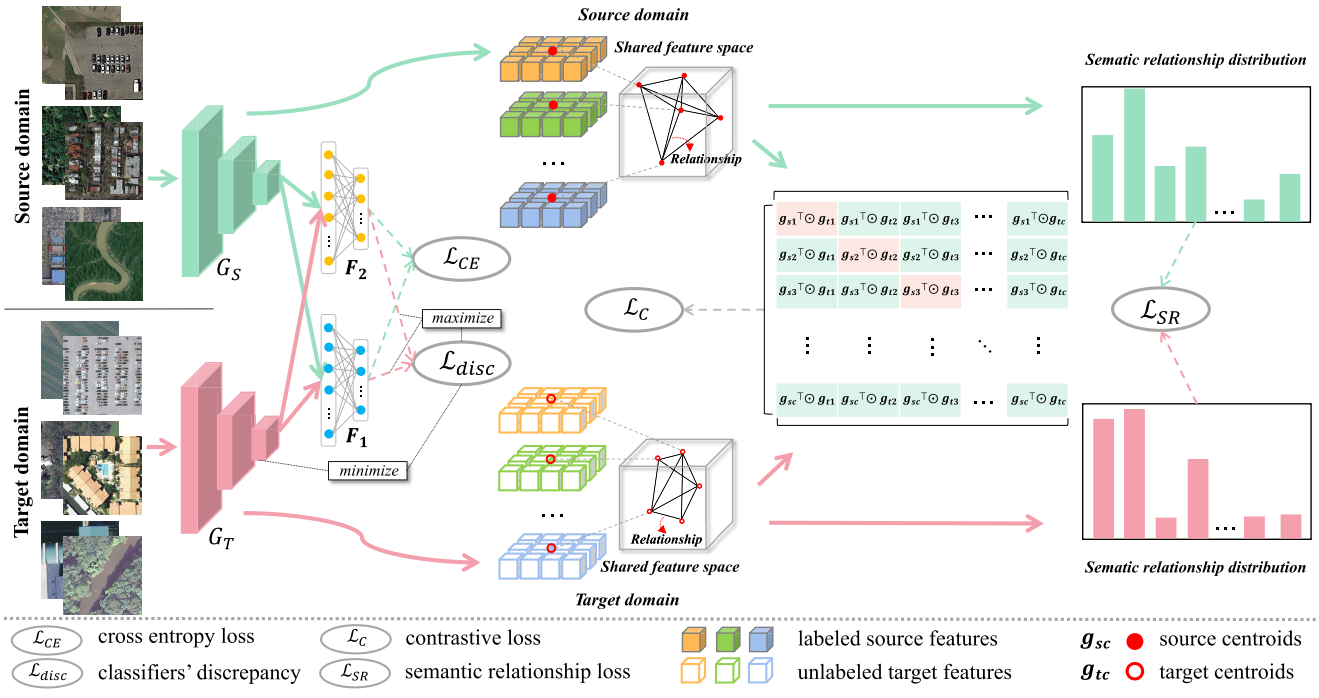


Fig. 2. Illustration of our proposed SRKT. Leveraging the domain-specific encoders: G_S for source domain and G_T for target domain, we project the data from two domains into a shared feature space to solve the issue of feature dimension discrepancy caused by different image scales in two datasets. Pseudo labeling technology assists in calculating semantic centers of target domain. Initially, for producing domain-invariant features across different domains, we adopt adversarial strategy through minimaxing the corresponding loss $\mathcal{L}_{\text{disk}}$. To achieve the alignment of conditional distributions between two domains, we conduct classwise prototype alignments by minimizing the corresponding loss $\mathcal{L}_C$. Moreover, to transfer the semantic relationships knowledge among categories underlying in source domain to target domain, we enforce the matching of semantic relationship distributions between two domains by minimizing the loss $\mathcal{L}_{\text{SR}}$, which can increase the class discriminability on the target domain.

semantic correlation knowledge. In detail, we introduce the overall process of our framework from three main aspects: domain-invariant features generating, discriminative representations deriving, and SRKT.

A. Preliminaries and Motivation

In the cross-domain scene recognition settings, there are two domains accessible: a labeled source domain and an unlabeled target domain. Concretely, we denote the source domain with n_s training instances by $\mathcal{D}_S = \{(\boldsymbol{x}_{si}, y_{si})\}_{i=1}^{n_s}$, where $\boldsymbol{x}_{si} \in \mathbb{R}^{d_s}$ is the ith source instance with d_s as its dimension and $y_{si} \in \{1, 2, \ldots, C\}$ is the corresponding label of $\boldsymbol{x}_{si}$ with C number of classes. As for unlabeled target domain $\mathcal{D}_T$, we denote it by $\mathcal{D}_T = \{(\boldsymbol{x}_{tj})\}_{j=1}^{n_t}$ with n_t unlabeled target instances, where $\boldsymbol{x}_{tj} \in \mathbb{R}^{d_t}$ is the jth unlabeled target instance with d_t-dimensional feature.

Considering there may exist image scale discrepancy among remote sensing datasets, we resize the images in source and target domain to the same scale before embedding. Then we construct feature encoders G_S and G_T for source domain and target domain to embed data from different domains into a common feature space. Specifically, we, respectively, denote the embedded features of source instance $\boldsymbol{x}_{si}$ and target instance $\boldsymbol{x}_{tj}$ in this shared feature space by $\boldsymbol{f}_{si} = G_S(\boldsymbol{x}_{si})$ and $\boldsymbol{f}_{tj} = G_T(\boldsymbol{x}_{tj})$, where $\boldsymbol{f}_{si}, \boldsymbol{f}_{tj} \in \mathbb{R}^{d_s}$ and d_s denotes the dimension of the shared feature space. Two shared but distinct classifiers F_1 and F_2 are constructed for making predictions for both domains. Relying on these classifiers, our ultimate goal is to correctly classify unlabeled target instances in the shared feature space.

In particular, since the resolution divergence and imaging condition difference caught by distinct remote sensors exhibit heterogeneities in cross-domain remote sensing images [22], [23], the existing domain shift will inevitably weaken the model transferability across different domains [11], [12], [20], [21]. Our final goal is to eliminate these heterogeneities as much as possible when conducting cross-domain remote sensing recognition tasks. To suppress this heterogeneity, SRKT emphasizes digging intrinsic semantic correlations in both domains for facilitating distinguished representations learning. We derive domain-invariant and discriminative features via an adversarial pattern between the domain-specific encoder and classifiers. By contrastive manipulation on categorywise prototypes, we narrow features' intraclass distance but drive interclass features further apart to yield more discriminative representations. Intuitively, inherent semantic structures can provide effective guidance for training models when the target is unlabeled. Inspired by this, we establish distributions for each class with semantic prototypes. Then we reduce the discrepancy between them to distill semantic relationship knowledge from the source to the target domain. The overview of our method is illustrated in Fig. 2.

To be concrete, we use feature encoder G_S for extracting the features of source domain $\boldsymbol{f}_{si} = G_S(\boldsymbol{x}_{si})$, and then utilize the classifiers F_1 and F_2 for obtaining the predictions of these features. The logits of $\boldsymbol{x}_{si}$ come from two classifiers are sent to softmax operation, and output the prediction

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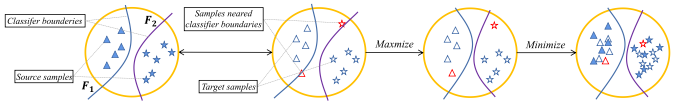


Fig. 3. Illustrative examples of detecting those samples in the target domain neared classifier boundaries.

probabilities $\boldsymbol{p}_{si}^1$, $\boldsymbol{p}_{si}^2$ of sample $\boldsymbol{x}_{si}$, where $\boldsymbol{p}_{si}^1$, $\boldsymbol{p}_{si}^2 \in \mathbb{R}^{d_c}$ and d_c denotes the dimension of the shared label space. For the sake of clarity, we directly denote two classifiers' prediction probabilities of $\boldsymbol{x}_{si}$ as $\boldsymbol{p}_{si}^1 = F_1(\boldsymbol{f}_{si})$ and $\boldsymbol{p}_{si}^2 = F_2(\boldsymbol{f}_{si})$, respectively. By exploiting ground truth y_s, we calculate the cross-entropy over source domain data $\boldsymbol{x}_{si}$ for model training. The cross-entropy loss formulates as follows:

$$\mathcal{L}_{\text{CE}} = -\sum_{i=1}^{n_s} \log \boldsymbol{p}_{si}^{1(y_{si})} + \log \boldsymbol{p}_{si}^{2(y_{si})} \tag{1}$$

where y_{si} is the ground truth of one source instance $\boldsymbol{x}_{si}$. Actually, calculating the cross-entropy loss over source data can provide valuable source supervision on the network.

B. Domain-Invariant Features Generating

To preliminarily undermine domain divergences existing in cross-domain remote sensing images, we train the network to yield domain-invariant and discriminative features across domains by an adversarial strategy. Two classifiers F_1 and F_2 are trained to find the samples in the target domain that are near the classification decision boundaries, and whose classification outputs are inconsistent. The illustrative example is shown in Fig. 3. Through directly enhancing this discrepancy of classifiers, we can pick up these instances and make one of the closest classifiers' boundaries include these instances for rectifying their classification results. Then utilizing G_T to generate features of target domain $\boldsymbol{f}_{tj} = G_T(\boldsymbol{x}_{tj})$ that can decrease the discrepancy of classifiers F_1 and F_2. As decreasing of the discrepancy, target instances that contain those instances near classifier boundaries will be driven far away from the closest decision boundaries. Due to the supervised training of the classifiers with the labeled source domain data throughout the whole training process, the classifier boundaries will keep the classifiers accurate on both source and target domain data in all training stages. Thus, the domain-invariant and discriminative features across domains could be obtained. The discrepancy between the two classifiers is computed with L_1 distance and is formulated as

$$\mathcal{L}_{\text{disk}} = \sum_{j=1}^{n_t} \left\| \boldsymbol{p}_{tj}^1 - \boldsymbol{p}_{tj}^2 \right\|_1 \tag{2}$$

where $\boldsymbol{p}_{tj}^1 = F_1(\boldsymbol{f}_{tj})$ and $\boldsymbol{p}_{tj}^2 = F_2(\boldsymbol{f}_{tj})$ denote the prediction probabilities output by F_1 and F_2, respectively. $\boldsymbol{p}_{tj}^1$ and $\boldsymbol{p}_{tj}^2$ are C-dimension prediction probability vectors of target domain data. Therefore, maximizing the discrepancy between classifiers F_1 and F_2 can be complemented by maximizing the value of $\mathcal{L}_{\text{disk}}$.

The minimizing process is another part of adversarial process, which can be conducted by exploiting the target encoder G_T to generate features that can minimize the value of $\mathcal{L}_{\text{disk}}$. Thus, we have constructed an adversarial process for model training. The minimaxing process can be written as

$$\min_{G_T} \max_{F_1, F_2} \sum_{j=1}^{n_t} \left\| \boldsymbol{p}_{tj}^1 - \boldsymbol{p}_{tj}^2 \right\|_1. \tag{3}$$

Here, depending on this minimaxing strategy, we can make target encoder G_T generate domain-invariant features across domains.

C. Discriminative Representations Deriving

For the sake of further promoting the generality of model and making features more discriminative, feature conditional distribution alignment is naturally considered. Contrastive alignment is an effective approach widely used in plenty of UDA Scenarios [66]. In this work, to conduct fine-grained feature conditional distribution alignment, we leverage the contrastive operation to narrow features' intraclass distance but drive interclass features further apart.

Nevertheless, the challenge of adopting a contrastive manner in remote sensing cross-scene recognition is how to construct the positive and negative samples due to there are no labels in the target domain. To tackle this problem, we permit classifiers trained with source data to assign confident pseudo labels for unlabeled target data $\boldsymbol{x}_t$. The pseudo labels assigning procedure can be represented as follows:

$$\begin{aligned} \hat{y}_{tj}^1 &= \arg\max_i \boldsymbol{p}_{tj}^{1(i)} \quad \boldsymbol{p}_{tj}^{1(i)} > \tau \\ \hat{y}_{tj}^2 &= \arg\max_i \boldsymbol{p}_{tj}^{2(i)} \quad \boldsymbol{p}_{tj}^{2(i)} > \tau \end{aligned} \tag{4}$$

where $\hat{y}_{tj}^1$ and $\hat{y}_{tj}^2$ are the two classifiers' pseudo labels of target domain instance $\boldsymbol{x}_{tj}$, respectively. The τ is the threshold used to select pseudo labels.

Then we calculate the source and target centroids based on the features in common feature space as classwise prototypes, also dubbed as semantic prototypes. Each prototype can be obtained by formulating as follows:

$$\begin{aligned} \boldsymbol{g}_{sc} &= \frac{1}{|\mathcal{D}_S^c|} \sum_{\boldsymbol{x}_{si} \in \mathcal{D}_S^c} G_S(\boldsymbol{x}_{si}) \\ \boldsymbol{g}_{tc} &= \frac{1}{|\mathcal{D}_T^c|} \sum_{\boldsymbol{x}_{tj} \in \mathcal{D}_T^c} G_T\left(\boldsymbol{x}_{tj}\right) \end{aligned} \tag{5}$$

where $\boldsymbol{g}_{sc}$ and $\boldsymbol{g}_{tc}$ are cth class prototypes in the source and target domain. And $|\mathcal{D}_S^c|$ and $|\mathcal{D}_T^c|$ are denoted as samples' number of cth class in source and target domains, respectively. After gaining the classwise prototypes, we can easily construct positive and negative sample pairs for contrastive learning. Specifically, we select one of the class prototypes in the source domain as an anchor, refer to the positive part, and select different class prototypes in the target domain as the negative part. We repeat this process until all categories of the source and target domains are traversed. This procedure

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can be formulated in terms of a contrastive learning loss as follows:

$$\mathcal{L}_C = \sum_{k=1}^{C} -\log \frac{e^{\boldsymbol{g}_{sk}^{\top} \odot \boldsymbol{g}_{tv}}}{\sum_{v \neq k}^{C} e^{\boldsymbol{g}_{sk}^{\top} \odot \boldsymbol{g}_{tv}}} \tag{6}$$

where dot product $\boldsymbol{g}_{sk}^{\top} \odot \boldsymbol{g}_{tv}$ is adopted for measuring the distance between the semantic prototype in source domain and the semantic prototype in target domain. It should be noted that semantic prototypes instead of instances are utilized for producing positive and negative sample pairs can significantly reduce computation overhead.

By minimizing this loss, feature conditional distribution alignment is accomplished. This further promotes the generality of model in the target domain and makes features more discriminative.

D. Semantic Relationship Knowledge Transfer

As can be seen in Fig. 2, in all domains, the "forest" element always appears in "river" scene images. Naturally, when seeing a "river," it is easy to associate it with scenes such as "river" and "forest," and rarely think about scenes such as "industrial" and "residential." Inspired by this, we are aware that inherent semantic structures can support guidance for training models when targets are unlabeled.

For the purpose of transmitting semantic correlation knowledge, we should first construct the semantic correlation structures in the respective domain. We calculate the similarity relationship between each class and other classes in their respective domains, generating two similarity distributions. Then we reduce the shift between distributions to transfer the semantic correlation knowledge from the source domain to the target domain. We define mean values $\boldsymbol{g}_{sk}$ and $\boldsymbol{g}_{sv}$ of features in each category of the source domain as the kth and vth semantic prototypes. Similarly, scenewise mean values $\boldsymbol{g}_{tk}$ and $\boldsymbol{g}_{tv}$ of features in the target domain denote the kth and vth semantic prototypes.

In each domain, dot product computations $\boldsymbol{g}_{sk}^{\top} \odot \boldsymbol{g}_{sv}$ and $\boldsymbol{g}_{tk}^{\top} \odot \boldsymbol{g}_{tv}$ are used for measuring the relevance strength among different scenes, e.g., "river" and "industrial." Based on this, we acquire the quantifiable semantic relationship information of each domain. The distinction between semantic relationship distributions can be formulated in terms of Jensen–Shannon divergence as follows:

$$\mathcal{L}_{\mathrm{SR}} = \frac{1}{2} \sum_{k=1}^{C} \sum_{v=1}^{C} \left(P_{kv}^{s} * \log \frac{P_{kv}^{s}}{\left(P_{kv}^{s} + P_{kv}^{t}\right)/2} + P_{kv}^{t} * \log \frac{P_{kv}^{t}}{\left(P_{kv}^{s} + P_{kv}^{t}\right)/2} \right) \tag{7}$$

where $P_{kv}^{s} = e^{\boldsymbol{g}_{sk}^{\top} \odot \boldsymbol{g}_{sv}}$ and $P_{kv}^{t} = e^{\boldsymbol{g}_{tk}^{\top} \odot \boldsymbol{g}_{tv}}$ are represented as the semantic relationship intensity between the kth class center in the source domain and vth class center in the target domain, respectively.

The migration of semantic correlation knowledge can constrain the semantic correlation distribution of the target domain data to obtain the semantic correlation knowledge on the source domain. Finally, it can make the target domain features obtain similar clear clustering distribution as the source domain features that are trained with ground truth, so as to promote the classification performance of the model on the target domain.

For the calculation of the semantic relationship between categories, we reuse the manner of computing semantic similarity which adopts the inner product between the class prototypes. This makes the computation of semantic relationship distribution simple and efficient. Furthermore, contrastive learning can make target domain semantic centers more discriminative and further yield more precise semantic correlation distribution in the target domain. The SRKT across domains helps the model to improve the recognizing ability in the target domain.

It is worth noting that each portion of our proposed method mutually advances others. For example, yielding domain-invariant features in an adversarial manner to eliminate domain shifts can lead to more accurate pseudo labels. High-quality pseudo labels will urge contrastive learning to better realize the fine-grained feature conditional distribution alignment across domains, enabling the network to produce more discriminative representations. Thus, SRKT from the source domain to the target domain also more effectively serves to prompt the generality of the model in the target domain. Conversely, effective SRKT prompts high-level adversarial learning and contrastive learning as well. All in all, the overall loss can be defined in terms of an SRKT loss as follows:

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \alpha \mathcal{L}_C + \beta \mathcal{L}_{\mathrm{SR}} \tag{8}$$

where α and β are the hyperparameters for trading off the balance of each component in the overall loss.

Simultaneously, the adversarial learning portion is incorporated into the training process by alternately maximizing and minimizing the loss $\mathcal{L}_{\mathrm{disk}}$, constituting a stepwise training strategy for generating domain-invariant but discriminative features. And the detailed algorithm of our proposed SRKT is presented in Algorithm 1.

IV. Experiment

In this section, we will introduce the popularly used domain adaptive remote sensing scene recognition datasets, our experimental settings, the quantitative and qualitative experimental results, and the detailed analysis of experimental results.

A. Datasets

To validate the effectiveness of SRKT, we select five widely used remote sensing scene recognition datasets for better realizing the experiments and displaying the results.

1) *UCM [36]:* The dataset contains 21 classes selected from aerial orthoimagery with a pixel resolution of 1 ft. 100 images measuring 256 × 256 pixels in each class.
2) *WHU-RS19 Dataset (WHU) [68]:* This dataset is constructed with satellite images exported from Google Earth, providing high-resolution up to 0.5 m. It contains 19 classes of meaningful scenes in high-resolution satellite imagery. For each category, there are about 50 samples that have a size of 600 × 600 pixels.

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Fig. 4. Samples in each dataset. (Top to bottom) Samples in WHU and UCM. The categories are agricultural, beach, building, forest, harbor, overpass, parking lot, river, and residential from the left to the right in each set of pictures. The differences and similarities between each dataset can be seen directly.

Algorithm 1 SRKT

Input: Source domain: $\mathcal{D}_S = \{(\boldsymbol{x}_{si}, y_{si})\}_{i=1}^{n_s}$; target domain: $\mathcal{D}_T = \{\boldsymbol{x}_{tj}\}_{j=1}^{n_t}$; batch size B; hyper-parameters: α, β.

Output: Feature encoders G_S, G_T and classifiers F_1, F_2.

1: Initialize the encoders: G_S, G_T and classifiers: F_1, F_2.
2: **for** $t = 1$ to *maximum-iteration* **do**
3: Randomly sample source batch $\{(\boldsymbol{x}_{si}, y_{si})\}_{i=1}^{B}$ and target batch $\{\boldsymbol{x}_{tj}\}_{j=1}^{B}$ from $\mathcal{D}_S$ and $\mathcal{D}_T$.
4: Obtain embedded features in common feature space: $\{\boldsymbol{f}_{si} = G_S(\boldsymbol{x}_{si})\}_{i=1}^{B}$, $\{\boldsymbol{f}_{tj} = \{G_T(\boldsymbol{x}_{tj})\}_{j=1}^{B}$.
5: Compute the cross-entropy loss on the source data by Eq. (1).
6: Calculate source and target centroids: $\boldsymbol{g}_{sc}$ and $\boldsymbol{g}_{tc}$ with pseudo labels by Eq. (5).
7: Calculate classwise similarity $e^{\boldsymbol{g}_{sk}^{\top} \odot \boldsymbol{g}_{tv}}$ and semantic relationship distributions by Eq. (6) and (7). Update the network by minimizing the loss in Eq. (8) with stochastic gradient descent (SGD).
8: Fix G_T and maximize discrepancy between F_1 and F_2. Update the network by maximizing the loss in Eq. (3) with SGD.
9: Fix F_1 and F_2 and generate features $\boldsymbol{f}_{tj} = G_T(\boldsymbol{x}_{tj})$ to minimize the discrepancy of F_1 and F_2. Update the network by minimizing the loss in Eq. (3) with SGD.
10: **end for**
11: Return feature encoders G_S, G_T and classifiers F_1, F_2.

3) *AID [35]:* This aerial image dataset is collected from Google Earth imagery. The dataset consists of 30 scenes. In all, the AID dataset has 10 000 images within 30 classes, and about 200–400 samples of size 600 × 600 pixels in each class.
4) *RSSCN7 Dataset (RSSCN7) [69]:* The dataset is collected from Google Earth which contains seven scene classes and 400 samples of size 400 × 400 pixels in each class, which are sampled on four different scales with 100 images per scale.
5) *NWPU-RESISC45 Dataset (NWPU) [70]:* This dataset is also extracted from Google Earth which is made up of 31 500 remote sensing images divided into 45 scene classes. Each category contains 700 images with a size of 256 × 256 pixels, and with a spatial resolution ranging from 0.2 to 30 m.

TABLE I
STATISTICS OF THE EVALUATED DATASETS

Datasets	# Instances	# Classes	# Tasks
UCM	900	9	U → W
WHU	477	9	W → U
UCM	700	6	U → A U → W U → R
AID	2220	6	A → U A → W A → R
WHU	316	6	W → U W → A W → R
RSSCN7	2400	6	R → U R → W R → A
UCM	1300	13	U → A
AID	4510	13	A → U
UCM	2000	20	U → N
NWPU	14000	20	N → U
AID	6280	19	A → N
NWPU	13300	19	N → A

B. Experiment Setting

For the sake of proving the effectiveness of our proposed method, we design three types of tasks with five datasets involving: AID, UCM, RSSCN7, WHU, and NWPU. The details of all datasets can be observed in Table I. We construct subdatasets by selecting some common scenes in these different datasets. The images in all datasets will be reshaped to the same size by 224 × 224 before training.

First, WHU and UCM datasets constitute the source and target domain. As we can see in Fig. 4, we select nine classes, including agricultural, beach, building, forest, harbor, overpass, parking lot, river, and residential, for experimenting on domain adaptive remote sensing scene recognition task: W→U. ResNet-101 [71] is used as the generator for extracting features.

Second, based on UCM, AID, and NWPU datasets, we experiment on six recognition tasks: U→A, A→U A→N, N→U, N→A, and U→N. ResNet-50 [71] is used as the generator for extracting features. Tasks A→U and U→A are built on 13 classes, including agriculture, airplane, baseball, beach, dense residential, forest, medium residential, parking, playground, port, river, sparse residential, and storage; As shown in Fig. 5, the U→N and N→U tasks involve 20 classes, including agricultural, airplane, baseball diamond, beach,

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TABLE II
RECOGNITION RESULTS (%) FOR WHU TO UCM TRANSFER RECOGNITION TASK WITH RESNET-101

Method / Classes	river	parking lot	overpass	harbor	forest	building	beach	residential	agricultural	Average
Source only	70.00	67.00	94.00	92.00	63.00	53.00	18.00	16.00	47.00	57.78
DANN [15]	79.00	84.00	89.00	94.00	59.00	62.00	15.00	20.00	54.00	61.78
PADA [56]	75.00	94.00	83.00	84.00	50.00	21.00	83.00	80.00	69.00	71.00
HAFN [67]	75.00	91.00	58.00	90.00	79.00	33.00	86.00	70.00	73.00	72.70
SAFN [67]	64.00	100.00	67.00	95.00	100.00	70.00	99.00	60.00	94.00	83.20
MCD [57]	79.00	96.00	87.00	66.00	99.00	68.00	94.00	73.00	93.00	85.44
SWD [47]	90.00	100.00	53.00	92.00	59.00	23.00	96.00	80.00	74.00	74.10
DTA [58]	87.00	76.00	89.00	91.00	91.00	62.00	96.00	76.00	78.00	82.90
DuEDL [24]	93.00	99.00	97.00	100.00	100.00	59.00	96.00	**95.00**	90.00	92.10
SRKT	**97.00**	**100.00**	**100.00**	**100.00**	**100.00**	**90.00**	**99.00**	84.00	**95.00**	**96.13**

chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium residential, overpass, mobile home park, parking lot, sparse residential, runway, river, storage tanks, and tennis court; The N→A and A→N tasks are constructed on 19 classes, including airport, baseball field, beach, bridge, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, parking, railway station, river, sparse residential, and storage tanks.

Third, making use of UCM, WHU, AID, and RSSCN7 datasets, we construct source and target domains through seeing one dataset as a domain. Fig. 6 displays instances of different categories in each dataset. Based on these datasets, 12 tasks are designed for displaying the superiority of our method, including U→W, U→A, U→R, W→U, W→A, W→R, A→U, A→W, A→R, R→U, R→W, and R→A. To make the domain adaptive remote sensing scene recognition experiments easy to carry out, we select the shared six categories of images from each dataset above, including farmland, forest, industrial, parking, residential, and river. Similarly, ResNet-50 is also used to extract features. Notably, to expand training data, we integrate dense residential and medium residential into the residential category in the UCM dataset.

The source and target feature extractors are separated parts and are all comprised of ResNet [71]. Two independent classifiers are all constructed with three liner layers. The stochastic gradient descent (SGD) is used for training feature extractors and classifiers. The momentum weight of 0.9 is used to train classifiers. We use a weight decay of 0.0005 and a learning rate of 0.001. The training epochs are 500, and the batch size is 64 for ResNet-101 [71] and 96 for ResNet-50 [71]. The hyperparameters α and β are set to 0.1 and 1. All experiments are run on GPU NVIDIA TITAN RTX with 24-GB memory. The PyTorch open-source framework and Python programming language are used to implement our codes.

C. Experiment Results

All experimental results are calculated in the same way, we run all cross-scene recognition tasks five times, and take the average value of the maximum value of each round as the final result for the corresponding task.

In WHU to UCM task, Table II shows that SRKT reaches the top accuracies on eight classes and obtains the highest average accuracy of 96.13%, which outperforms the best current method DuEDL by 4.03%. In this task, we select domain-adversarial neural network (DANN) [15], PADA [56], hard adaptive feature norm (HAFN) [67], stepwise adaptive feature norm (SAFN) [67], maximum classifier discrepancy (MCD) [57], SWD [47], DTA [58], and DuEDL [24] for comparison. This task preliminarily exhibits the effectiveness of reducing the heterogeneity from satellite images to aerial images and outstanding performance in cross-sensor remote scene sensing recognition. At the same time, the accuracy of each category is explicitly listed for illustrating the recognition performance of all methods on the UCM dataset.

For the second group of tasks, we adopt deep domain confusion (DDC) [37], DAN [40], Deep CORAL [39], DANN [15], and the current most advanced method DATSNET [26] for comparison. As we can see in Table III, our proposed SRKT reaches the best accuracies on four tasks and acquires the second-highest accuracies of the rest two tasks. We compute the average accuracy of all tasks to systemically describe the improvement of overall mean accuracy. Notably, SRKT obtains the highest average accuracy of 91.20%, which outperforms DATSNET by 3.52%, which effectively proves the ability of cross-sensor scene recognition.

Comparison methods DDC [37], Deep CORAL [39], DAN [38], RevGrad [14], joint adaptation networks (JAN) [9], MRAN [46], and AMRAN [1] are chosen to compare with our method in the third group of tasks. Table IV shows that our SRKT achieves the highest accuracies on almost all tasks, besides A→U and A→W tasks. We also calculate the mean accuracy of all tasks to comprehensively describe the performance growth brought by our method on all tasks. SRKT reaches the highest average accuracy of 90.66%, which outperforms the current best method AMRAN by 3.42%. Based on excellent results on these extensive and comprehensive tasks, the power of our approach in mitigating domain divergence and generalizing model is obvious.

D. Empirical Analysis

In this part, we conduct several empirical analyses to verify the effectiveness and robustness of our SRKT.

1) Ablation Study: A series of ablation studies are completed for analyzing the effect of every major component in our SRKT. By setting the corresponding loss to 0 in (3) and (8), we can realize the effect of removing this module. So we design five experiments, including only supervised source data

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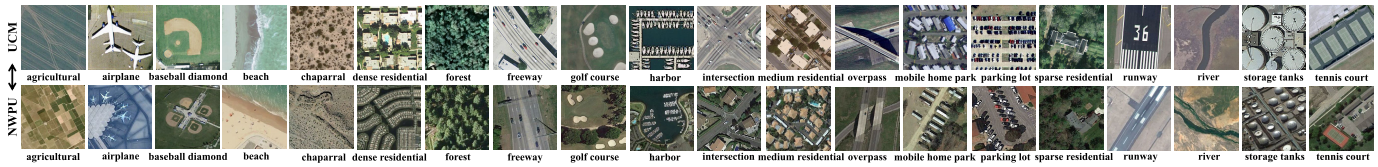


Fig. 5. Samples in each dataset are shown. (Top to bottom) Samples in UCM and NWPU. The categories are agricultural, airplane, baseball diamond, beach, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium residential, overpass, mobile home park, parking lot, sparse residential, runway, river, storage tanks, and tennis court from the left to the right in each set of pictures.

TABLE III
RECOGNITION RESULTS (%) FOR CROSS-DOMAIN SCENE RECOGNITION (UCM, AID, AND NWPU) WITH RESNET-50

Method / Task	A → U	U → A	N → U	U → N	N → A	A → N	Average
Source only	62.34	61.82	70.20	61.55	88.21	74.53	69.78
DDC [37]	68.38	52.84	80.47	52.12	85.30	81.31	70.07
DAN [40]	75.40	56.63	85.27	54.15	86.66	86.72	74.14
Deep CORAL [39]	65.69	53.94	79.75	52.21	85.10	81.40	69.68
DANN [15]	67.21	52.84	84.40	53.31	86.08	82.27	71.02
DATSNET [26]	85.67	88.40	**91.87**	77.53	94.33	**88.40**	87.68
SRKT	**95.38**	**91.35**	91.75	**86.40**	**94.97**	87.04	**91.20**

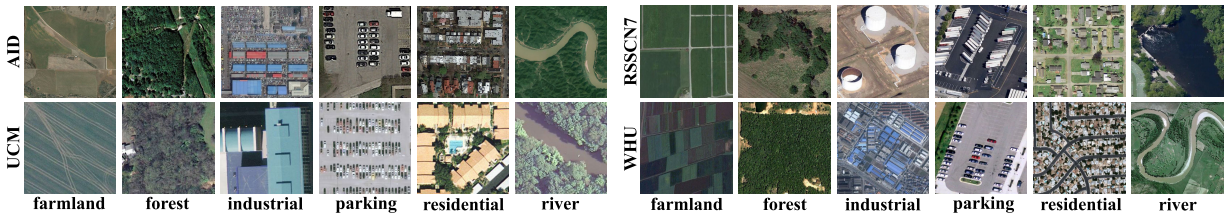


Fig. 6. Samples in each dataset are shown. (Top left to bottom right) Samples in AID, samples in UCM, samples in RSSCN7, and samples in WHU. The categories are farmland, forest, industrial, parking, residential, and river from the left to the right in each set of images.

TABLE IV
RECOGNITION RESULTS (%) FOR CROSS-DOMAIN SCENE RECOGNITION (UCM, WHU, AID, AND RSSCN7) WITH RESNET-50

Method / Task	U→W	U→A	U→R	W→U	W→A	W→R	A→U	A→W	A→R	R →U	R→W	R→A	Average
Source only	71.20	66.44	56.50	75.14	93.11	71.25	74.86	98.42	76.88	72.14	86.71	85.18	77.32
DDC [37]	71.84	66.26	57.71	78.00	90.09	70.13	78.29	99.21	75.63	79.71	89.24	84.68	78.40
Deep CORAL [39]	73.73	65.95	56.46	79.43	92.61	72.79	79.29	99.37	76.96	83.43	89.87	85.45	79.61
DAN [38]	74.68	67.07	57.50	80.14	92.79	72.79	80.29	99.37	76.88	82.86	88.92	85.32	79.88
RevGrad [14]	72.47	67.48	55.83	80.86	92.30	72.75	80.71	99.05	76.83	83.86	90.19	85.50	79.82
JAN [9]	76.58	69.19	61.89	81.86	92.29	73.85	82.86	98.10	78.13	81.57	88.61	84.28	80.75
MRAN [46]	78.32	82.99	64.78	81.71	94.06	79.16	84.71	98.73	78.96	83.06	93.35	93.87	84.47
AMRAN [1]	89.56	84.80	69.02	85.20	94.82	79.50	**88.16**	**99.68**	81.00	87.56	94.59	94.48	87.36
SRKT	**93.98**	**95.27**	**74.53**	**88.34**	**97.01**	**85.55**	84.83	98.73	**85.34**	**89.17**	**96.77**	**98.41**	**90.66**

training, deleting adversarial operation, deleting contrastive learning, deleting semantic relationship transferring module, and fully adopting all components.

The detailed results of the ablation studies can be seen in Table V. Obviously, each component plays an important role in the adaptation process. Using only source data to train the model reaches the lowest accuracy. The contrastive learning component plays the most important role among all parts with the highest improvement in recognition accuracy. Adversarial learning and SRKT parts all contribute to elevating the performance of recognition. Finally, the organic combination of these three ingredients promotes the recognition accuracy to the highest, which explicitly demonstrates that the main components of our proposed method play a critical role in eliminating cross-scene differences.

To sufficiently testify to the contribution brought by each module, we unify each module with the source model. Detailed results can be seen in Table VI. Comparing the results with the model only trained by source data, we can observe that every module can prompt the classification precision of the model by alone integrating each module with the cross-entropy loss on source data. It is consistent with the results shown in Table V. Combining adversarial learning and SRKT parts separately with the cross-entropy loss on source data all contribute to elevating the performance of recognition, which outperforms the source model by 4.54% and 4.14%, respectively. The contrastive learning component brings the most significant performance ascending among all parts on scene recognition accuracy, obtaining a 10.42% accuracy improvement.

2) Feature Visualization: To visually describe the improvement of scene recognition accuracy, we conduct qualitative t-Distributed stochastic neighbor embedding (t-SNE) experiments on the WHU to UCM transfer task to visualize the change brought by the proposed SRKT. In Fig. 7, every

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TABLE V
ABLATION STUDY OF SRKT (UCM, WHU, AID, AND RSSCN7) WITH RESNET-50

Component / Task	U→W	U→A	U→ R	W→U	W→A	W→R	A→U	A→W	A →R	R→U	R→W	R→A	Average
Source only	71.20	66.44	56.50	75.14	93.11	71.25	74.86	98.42	76.88	72.14	86.71	85.18	77.32
SRKT (w/o L_{disc})	93.73	94.98	70.18	85.74	96.12	81.22	82.86	97.66	81.68	88.89	95.76	98.33	88.93
SRKT (w/o L_C)	86.71	85.58	67.15	85.14	94.50	77.71	81.83	98.04	83.79	84.86	92.15	95.57	86.09
SRKT (w/o L_{SR})	93.16	94.05	70.62	86.51	96.50	82.32	83.29	98.42	83.97	88.29	95.56	98.22	89.24
SRKT	**93.98**	**95.27**	**74.53**	**88.34**	**97.01**	**85.55**	**84.83**	**98.73**	**85.34**	**89.17**	**96.77**	**98.41**	**90.66**

TABLE VI
ABLATION STUDY OF SRKT (UCM, WHU, AID, AND RSSCN7) WITH RESNET-50

Component				Task												
L_{CE}	L_{disc}	L_{SR}	L_C	U → W	U → A	U → R	W → U	W → A	W → R	A → U	A → W	A → R	R → U	R → W	R → A	Average
✓				71.20	66.44	56.50	75.14	93.11	71.25	74.86	98.42	76.88	72.14	86.71	85.18	77.32
✓	✓			75.11	75.15	63.35	81.00	94.12	79.50	78.64	97.78	81.54	76.21	88.13	91.72	81.86
✓		✓		77.58	76.12	65.16	81.50	93.83	74.48	76.14	98.31	79.13	76.71	89.77	88.81	81.46
✓			✓	91.30	92.64	69.46	85.57	95.24	81.48	82.00	98.42	82.04	86.43	93.04	95.31	87.74
✓	✓	✓	✓	**93.99**	**95.27**	**74.53**	**88.34**	**97.01**	**85.55**	**84.83**	**98.73**	**85.34**	**89.17**	**96.77**	**98.41**	**90.66**

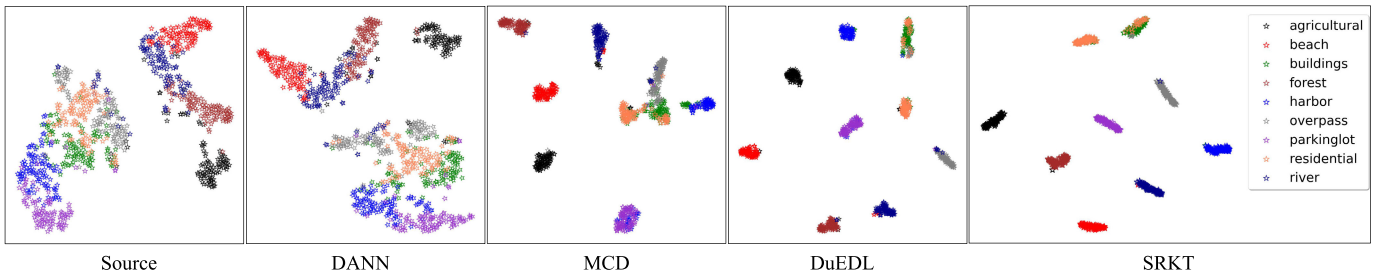


Fig. 7. Feature visualization of UCM dataset. (Left to right) Qualitative experiment (t-SNE) results of source only trained model, DANN, MCD, DuEDL, and the proposed SRKT on WHU to UCM transfer task.

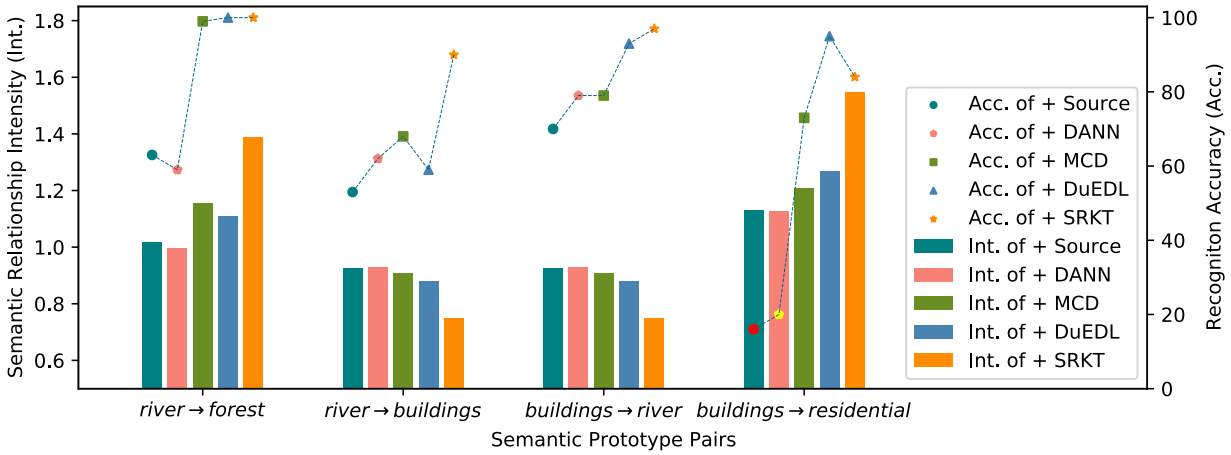


Fig. 8. Comparisons of the semantic relationship intensity (Int.) between scenes in target domain and the recognition accuracy (Acc.) on target domain for different methods.

star symbol represents the feature of an instance in the target domain, and different colors mean different categories. Detailed recognition accuracies in each scene refer to Table II. Fig. 7 distinctly shows the visual results on source only trained model [71], DANN [15], MCD [57], DuEDL [24], and our proposed SRKT, which indicate that SRKT obtains the least mismatch and the best clustering effect among these classical methods. SRKT appears obvious mismatch between building and residential categories. This is because the buildings and residential categories own amounts of building contents, and the appearance of these two classes is too similar. Despite all this, our proposed SRKT still reduces the classification errors in these two scenes to the lowest compared with other methods.

3) Semantic Relationship Intensity: To further show the association between remote sensing cross-scene recognition performance and semantic relationship intensity $e^{\boldsymbol{g}_{sk}^{\top}\odot\boldsymbol{g}_{sv}}$ in (7), we list the values of semantic relationship in four scene pairs selected from in WHU→UCM cross-scene recognition task. Combined with Table II, we compare the recognition accuracies of the posterior scene in each scene pair of all methods, i.e., Source only [71], DANN [15], MCD [57],

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DuEDL [24], and SRKT. In Fig. 8, we can see that the recognition accuracies rise when semantic relationship intensities between related scenes increase, e.g., river→forest and buildings→residential scene pairs display that with the semantic relationship increasing, the recognition performances obviously enhance. Note that recognition accuracies of DuEDL and SRKT in buildings→residential appear to decrease with the semantic relationship intensity arising. This phenomenon is because the appearances of these two classes are overly alike. Despite all this, the overall recognition accuracy on both buildings and residential scenes is significantly higher when viewed together. On the other hand, the river→buildings and buildings→river pairs exhibit with the semantic relationship between irrelevant scenes decreasing, the recognition performances also acquire a significant growth.

V. Conclusion

In this article, we present a novel unsupervised SRKT framework for domain adaptive remote sensing scene recognition. The SRKT successfully uses adversarial behavior between the target generator and classifiers to emerge domain-invariant but discriminative features, effectively enhancing the recognizing capability of the model on the target domain. By contrastive operation, more discriminative representations are acquired. Most importantly, our proposed network reveals semantic relationship structures underlying the target domain by matching the semantic relationship distributions between the two domains. Depending on this, SRKT facilitates the model to create more distinguishable features in the target domain. Ultimately, we conduct quantitative and qualitative experiments on five popularly used remote sensing datasets. The results of the quantitative experiment and empirical analysis confirm SRKT's superior performance in domain adaptive remote sensing scene recognition tasks.

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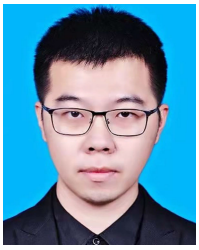
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